

# CTMT Null Leakage and Instability

Covariance-Induced Open Sectors, Amplification Modes, and Operational Collapse

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July 6, 2026

## Abstract

Chronotopic Metric Theory (CTMT) reconstructs local observable transport through the resolution geometry

$$\mathcal{G} = (O; R, N, \Sigma),$$

where  $R = \text{Im}(J)$  is the resolved sector,  $N = \text{Ker}(J^T)$  is the unresolved observable sector, and  $\Sigma$  is empirical covariance. Earlier CTMT layers establish admissible morphisms, collapse, holonomy, and global similarity atlases. This paper develops the next foundational mechanism: *null leakage and instability*. Null leakage is the operational phenomenon by which unresolved covariance components influence resolved reconstruction through the resolved–null coupling block

$$C_{RN} = P_R \Sigma P_N.$$

It is not postulated as a physical force, hidden variable, or metaphysical channel. It is a measurable obstruction inside the observable covariance geometry.

The central result is a leakage–instability lock. Nonzero coupling does not by itself imply instability. It shrinks admissible automorphisms and creates a channel through which perturbations of the null sector may amplify resolved conditioning, Fisher response, holonomy error, and atlas gluing defects. A system is operationally stable only when rank, projectors, covariance blocks, coupling, response metric, and loop transports remain bounded under the chosen protocol. Instability occurs when null leakage produces unbounded or out-of-tolerance defects in this hierarchy.

The paper provides definitions, theorem-level statements, proof sketches, falsification batteries, counterexamples, health bars, and reproducibility appendices. Its scope is deliberately operational: CTMT null leakage is a property of observable resolution geometry and covariance estimation. Physical interpretation remains an open problem.

## Contents

|          |                                    |          |
|----------|------------------------------------|----------|
| <b>1</b> | <b>Introduction</b>                | <b>2</b> |
| <b>2</b> | <b>Local Leakage Geometry</b>      | <b>2</b> |
| <b>3</b> | <b>Admissibility Under Leakage</b> | <b>3</b> |
| <b>4</b> | <b>Instability</b>                 | <b>4</b> |
| <b>5</b> | <b>Leakage–Instability Lock</b>    | <b>4</b> |
| <b>6</b> | <b>Fisher Amplification</b>        | <b>5</b> |
| <b>7</b> | <b>Holonomy Under Leakage</b>      | <b>5</b> |

|                                                  |                   |
|--------------------------------------------------|-------------------|
| <a href="#">8 Atlas Instability</a>              | <a href="#">5</a> |
| <a href="#">9 Counterexamples</a>                | <a href="#">6</a> |
| <a href="#">10 Numerical Battery</a>             | <a href="#">6</a> |
| <a href="#">11 Master Health Table</a>           | <a href="#">7</a> |
| <a href="#">12 Non-Claims and Limitations</a>    | <a href="#">7</a> |
| <a href="#">13 Open Problems</a>                 | <a href="#">7</a> |
| <a href="#">14 Conclusion</a>                    | <a href="#">8</a> |
| <a href="#">A Appendix A: JSON Output Schema</a> | <a href="#">8</a> |
| <a href="#">B Appendix B: Minimal Pseudocode</a> | <a href="#">8</a> |

## 1 Introduction

The CTMT resolution geometry

$$\mathcal{G} = (O; R, N, \Sigma)$$

separates observable directions into a resolved sector  $R$  and an unresolved sector  $N$ . This split is induced by the local response operator  $J$  through

$$R = \text{Im}(J), \quad N = \text{Ker}(J^T).$$

The covariance operator then decomposes in an adapted basis  $Q = [U_R \ U_N]$  as

$$Q^T \Sigma Q = \begin{pmatrix} \Sigma_R & C_{RN} \\ C_{RN}^T & \Sigma_N \end{pmatrix}.$$

The off-diagonal block  $C_{RN}$  is the central object of this paper.

If  $C_{RN} = 0$ , observation covariance is block diagonal relative to the resolved–null split. Resolved uncertainty and unresolved uncertainty are statistically decoupled at the level of the chosen observable protocol. If  $C_{RN} \neq 0$ , the covariance geometry does not respect the response-induced sector split. This mismatch is what CTMT calls *null leakage*.

The phrase must be read carefully. Null leakage does not mean that a hidden physical substance flows from  $N$  into  $R$ . It means that unresolved observable covariance affects resolved reconstruction through a measurable cross-sector block. The null sector is therefore not disposable noise. It can constrain admissible morphisms, shrink stabilizers, deform Fisher information, perturb holonomy, and destabilize atlas gluing.

## 2 Local Leakage Geometry

**Definition 2.1** (Resolved–null covariance coupling). Let  $P_R$  and  $P_N$  be the orthogonal projectors onto  $R$  and  $N$ . The resolved–null covariance coupling operator is

$$C_{RN} = P_R \Sigma P_N.$$

In an adapted basis, it is the off-diagonal covariance block in

$$\Sigma = \begin{pmatrix} \Sigma_R & C_{RN} \\ C_{RN}^T & \Sigma_N \end{pmatrix}.$$

**Definition 2.2** (Null leakage). Null leakage occurs whenever

$$C_{RN} \neq 0.$$

Equivalently, the covariance operator fails to preserve the sector decomposition  $O = R \oplus N$ .

**Proposition 2.3** (Coupling criterion). *The following are equivalent:*

- (i)  $C_{RN} = 0$ ;
- (ii)  $P_R \Sigma P_N = 0$ ;
- (iii)  $\Sigma(R) \subseteq R$  and  $\Sigma(N) \subseteq N$ ;
- (iv) *covariance is block diagonal in an adapted resolved-null basis.*

*Proof.* If  $C_{RN} = P_R \Sigma P_N = 0$  and symmetry gives  $P_N \Sigma P_R = 0$ , then  $\Sigma$  has no cross-sector block and preserves both subspaces. Conversely, if  $\Sigma$  preserves both  $R$  and  $N$ , the off-diagonal projections vanish.  $\square$

**Definition 2.4** (Leakage amplitude). The scalar leakage amplitude is

$$\lambda_{RN} = \|C_{RN}\|_F.$$

A normalized leakage ratio is

$$\rho_{RN} = \frac{\|C_{RN}\|_F}{\sqrt{\|\Sigma_R\|_F \|\Sigma_N\|_F} + \epsilon},$$

where  $\epsilon > 0$  is a numerical safeguard.

**Remark 2.5.**  $\lambda_{RN}$  is scale-dependent, while  $\rho_{RN}$  is more useful for comparing experiments with different measurement units or covariance magnitudes.

### 3 Admissibility Under Leakage

**Definition 3.1** (Admissible automorphism). A linearized map  $A : O \rightarrow O$  is admissible for  $\mathcal{G}$  if

$$A(R) = R, \quad A(N) = N, \quad A^T \Sigma A = \Sigma.$$

**Theorem 3.2** (Leakage shrinks automorphisms). *If  $C_{RN} \neq 0$ , then the admissible automorphism group is constrained by the stabilizer of the coupling block. In adapted block form, if*

$$A = \begin{pmatrix} A_R & 0 \\ 0 & A_N \end{pmatrix},$$

*then admissibility requires*

$$A_R^T \Sigma_R A_R = \Sigma_R, \quad A_N^T \Sigma_N A_N = \Sigma_N, \quad A_R^T C_{RN} A_N = C_{RN}.$$

*Thus nonzero leakage removes transformations that would be allowed by the resolved and null diagonal blocks alone.*

*Proof.* Substitute the block forms of  $A$  and  $\Sigma$  into  $A^T \Sigma A = \Sigma$  and compare diagonal and off-diagonal blocks. The off-diagonal equality is exactly  $A_R^T C_{RN} A_N = C_{RN}$ . When  $C_{RN} = 0$ , this constraint vanishes. When  $C_{RN} \neq 0$ , admissible pairs  $(A_R, A_N)$  must stabilize the coupling block.  $\square$

**Corollary 3.3** (Leakage rigidity). *Null leakage increases geometric rigidity. It does not necessarily collapse the system, but it reduces the dimension or freedom of admissible transformations unless the coupling block is degenerate in a compatible way.*

## 4 Instability

**Definition 4.1** (Instability defect vector). For a transported or perturbed geometry, define the instability defect vector

$$\Delta = (\delta_{\text{rank}}, \delta_R, \delta_N, \delta_\Sigma, \delta_{RN}, \delta_F, \delta_H, \kappa),$$

where

$$\delta_{\text{rank}} = |\text{rank}(J') - \text{rank}(J)|, \quad (1)$$

$$\delta_R = \|P'_R - P_R\|, \quad (2)$$

$$\delta_N = \|P'_N - P_N\|, \quad (3)$$

$$\delta_\Sigma = \|\Sigma' - \Sigma\|, \quad (4)$$

$$\delta_{RN} = \|C'_{RN} - C_{RN}\|, \quad (5)$$

$$\delta_F = \|F' - F\|, \quad (6)$$

$$\delta_H = \text{dist}(H_\gamma, \text{Aut}(\mathcal{G})), \quad (7)$$

$$\kappa = \text{cond}(H_\gamma). \quad (8)$$

**Definition 4.2** (Operational instability). A CTMT chart or chart family is operationally unstable if one or more required components of  $\Delta$  exceeds the specified tolerance vector. Instability is therefore protocol-relative and observable.

**Definition 4.3** (Leakage-driven instability). Instability is leakage-driven when the first exceeded health bar is downstream of  $C_{RN}$ : coupling defect, Fisher amplification, stabilizer shrinkage, holonomy defect, or atlas gluing failure, while rank may still remain preserved.

**Principle 4.4** (Rank is not stability). *Rank preservation is only the first health bar. A system may preserve rank and still be unstable through leakage, covariance deformation, Fisher blow-up, stabilizer discontinuity, or holonomy collapse.*

## 5 Leakage–Instability Lock

**Theorem 5.1** (Leakage is necessary for cross-sector covariance instability). *Assume  $R$  and  $N$  are fixed and covariance perturbations are the only source of instability. If  $C_{RN} = 0$  and perturbations preserve block diagonality, then there is no cross-sector covariance instability. Conversely, any instability whose first-order term is an off-diagonal resolved–null covariance defect has nonzero first-order leakage.*

*Proof.* With fixed sectors, cross-sector covariance is exactly the off-diagonal block  $C_{RN} = P_R \Sigma P_N$ . If this block remains zero under perturbation, no resolved–null covariance channel exists. If the first-order instability is cross-sector, then the first variation  $P_R \dot{\Sigma} P_N$  is nonzero, which is precisely first-order leakage.  $\square$

**Theorem 5.2** (Leakage–instability lock). *Let a CTMT protocol require bounded rank, sector projectors, covariance blocks, Fisher response, coupling, and holonomy. Then:*

- (a) *Nonzero  $C_{RN}$  alone does not imply instability.*
- (b) *Nonzero  $C_{RN}$  creates an admissibility constraint  $A_R^T C_{RN} A_N = C_{RN}$ .*
- (c) *If perturbations of  $N$  amplify  $\delta_{RN}$ ,  $\delta_F$ ,  $\delta_H$ , or holonomy conditioning beyond tolerance, the system is unstable even when rank is preserved.*
- (d) *Therefore leakage becomes instability precisely when it produces a protocol-level closure defect.*

*Proof.* Part (a) follows because geometries with fixed nonzero  $C_{RN}$  may still have well-defined automorphism groups and bounded holonomy. Part (b) is the block admissibility theorem. Part (c) follows from the definition of operational instability. Part (d) combines these statements: leakage is a structural channel, while instability is the failure of bounded closure through that channel.  $\square$

**Corollary 5.3** (Survival with leakage). *A CTMT system can be open, coupled, and still stable. Stability requires not zero leakage, but bounded leakage compatible with the admissible geometry.*

## 6 Fisher Amplification

The Fisher-type response metric is

$$F = J^T \Sigma^{-1} J.$$

When  $\Sigma$  has resolved-null coupling, inversion mixes the blocks through Schur complements.

**Proposition 6.1** (Schur complement leakage). *Assume  $\Sigma_N$  is invertible. The effective resolved covariance entering the block inverse is*

$$S_R = \Sigma_R - C_{RN} \Sigma_N^{-1} C_{RN}^T.$$

*Thus null covariance and coupling modify the resolved inverse geometry even if  $J$  has image entirely in  $R$ .*

*Proof.* This is the standard block inverse formula. The resolved block of  $\Sigma^{-1}$  depends on the Schur complement  $S_R^{-1}$ , which contains the term  $C_{RN} \Sigma_N^{-1} C_{RN}^T$ .  $\square$

**Corollary 6.2** (Fisher instability criterion). *If leakage drives  $S_R$  toward singularity, then  $F$  may become ill-conditioned or unstable even when  $J$  and  $\text{rank}(J)$  remain unchanged.*

## 7 Holonomy Under Leakage

**Definition 7.1** (Leakage-sensitive holonomy). A holonomy loop  $H_\gamma$  is leakage-sensitive if its admissibility defect changes at first order under perturbations of  $C_{RN}$ :

$$\left. \frac{d}{dt} \right|_{t=0} \text{dist}(H_\gamma(t), \text{Aut}(\mathcal{G}(t))) \neq 0.$$

**Proposition 7.2** (Coupling constrains loop products). *For a loop product  $H_\gamma$  to remain admissible in a coupled geometry, its resolved and null blocks must stabilize  $C_{RN}$ :*

$$H_R^T C_{RN} H_N = C_{RN}.$$

*Failure of this equality is a holonomy-level leakage defect.*

*Proof.* A loop product is admissible exactly when it preserves the full covariance block matrix and sectors. Comparing the off-diagonal block gives the stated condition.  $\square$

## 8 Atlas Instability

In a Global Similarity Atlas, each chart has its own coupling block  $C_{RN,\alpha}$ . Leakage becomes a global issue when these blocks cannot be transported consistently on overlaps.

**Definition 8.1** (Atlas leakage obstruction). Given a transition candidate  $T_{\alpha\beta}$ , the atlas leakage obstruction is

$$\delta_{RN}^{\alpha\beta} = \|C_{RN,\alpha}(T_{\alpha\beta}) - C_{RN,\beta}\|.$$

If this exceeds tolerance, the overlap fails coupling compatibility.

**Theorem 8.2** (Leakage atlas obstruction). *A family of charts with equal rank and compatible diagonal covariance spectra may still fail to form a Global Similarity Atlas if the coupling blocks cannot be matched under admissible transitions.*

*Proof.* Global atlas gluing requires preservation of the full resolution type, including  $C_{RN}$ . Equal rank and diagonal block spectra do not determine the coupling block. Therefore incompatible coupling blocks obstruct atlas gluing.  $\square$

## 9 Counterexamples

**Example 9.1** (Leakage without instability). Let  $\Sigma_R = I$ ,  $\Sigma_N = 1$ , and  $C_{RN} = ce_1$  with small fixed  $c \neq 0$ . If all tested transitions stabilize  $e_1$  and all closure defects remain below tolerance, the chart is coupled but stable.

**Example 9.2** (Rank stable but Fisher unstable). Keep  $J$  fixed and vary  $C_{RN}$  so that

$$S_R = \Sigma_R - C_{RN}\Sigma_N^{-1}C_{RN}^T$$

approaches singularity. Rank is unchanged, but  $F = J^T\Sigma^{-1}J$  becomes ill-conditioned.

**Example 9.3** (Null inflation). Let  $A_N = \lambda I_N$  with  $\lambda \neq 1$ . Sectors and rank may survive, but  $A_N^T\Sigma_N A_N$  and  $A_R^T C_{RN} A_N$  generally fail covariance and coupling closure.

**Example 9.4** (Holonomy leakage collapse). A loop product may preserve  $R$  and  $N$  but rotate the coupling vector  $C_{RN}$  into a different direction. Then the loop is sector-preserving but not covariance-admissible. This is holonomy-level leakage collapse.

## 10 Numerical Battery

A reproducible leakage battery should report the following sequence.

- Step 1.** Estimate  $J$  and compute  $R, N, P_R, P_N$ .
- Step 2.** Estimate empirical covariance  $\Sigma$ .
- Step 3.** Compute adapted blocks  $\Sigma_R, C_{RN}, \Sigma_N$ .
- Step 4.** Report  $\lambda_{RN} = \|C_{RN}\|_F$  and normalized  $\rho_{RN}$ .
- Step 5.** Compute Schur complement  $S_R = \Sigma_R - C_{RN}\Sigma_N^{-1}C_{RN}^T$ .
- Step 6.** Report  $\text{cond}(S_R)$  and  $\text{cond}(F)$ .
- Step 7.** Test identity, sector rotations, resolved-null mixing, null inflation, rank collapse, covariance anisotropy, and holonomy loops.
- Step 8.** For every test emit rank, projector, covariance, coupling, Fisher, holonomy, and conditioning defects.
- Step 9.** Declare instability only from the defect table.

## 11 Master Health Table

Table 1: Null leakage and instability diagnostics.

| Diagnostic              | Construction                        | Instability meaning                     |
|-------------------------|-------------------------------------|-----------------------------------------|
| Rank                    | $\text{rank}(J)$                    | resolved dimension loss                 |
| Projectors              | $P_R, P_N$                          | sector drift                            |
| Leakage amplitude       | $\ C_{RN}\ _F$                      | cross-sector covariance strength        |
| Normalized leakage      | $\rho_{RN}$                         | scale-adjusted leakage                  |
| Schur complement        | $S_R$                               | resolved covariance after null coupling |
| Fisher conditioning     | $\text{cond}(J^T \Sigma^{-1} J)$    | distinguishability amplification        |
| Automorphism shrinkage  | stabilizer of $C_{RN}$              | reduced admissible freedom              |
| Holonomy coupling error | $\ H_R^T C_{RN} H_N - C_{RN}\ $     | loop leakage defect                     |
| Atlas leakage error     | overlap $\delta_{RN}^{\alpha\beta}$ | gluing obstruction                      |

## 12 Non-Claims and Limitations

1. Null leakage is not claimed to be a physical substance, force, or hidden causal channel.
2. Nonzero leakage is not automatically instability.
3. Zero leakage in one observational protocol does not prove physical closure; it only says that the measured covariance is block diagonal relative to that protocol.
4. Rank stability is not sufficient for stability.
5. Fisher blow-up may reflect covariance estimation problems, insufficient sampling, or ridge choice; it is not automatically physical collapse.
6. Thresholds must be stated before falsification. Post-hoc tolerance selection weakens the test.

## 13 Open Problems

**Open Problem 13.1** (Sharp leakage stability bound). Find sufficient conditions on  $\|C_{RN}\|$ ,  $\Sigma_R$ , and  $\Sigma_N$  ensuring bounded  $\text{cond}(S_R)$  and bounded Fisher response.

**Open Problem 13.2** (Leakage normal forms). Classify  $C_{RN}$  under covariance-preserving sector automorphisms.

**Open Problem 13.3** (Statistical leakage thresholds). Develop finite-sample confidence intervals for  $\lambda_{RN}$  and  $\rho_{RN}$  under repeated observation noise.

**Open Problem 13.4** (Holonomy leakage classification). Classify loop products that preserve sectors but fail coupling stabilization.

**Open Problem 13.5** (Physical realization). Determine when a measured nonzero  $C_{RN}$  corresponds to sensor layout, bandwidth, unresolved dynamics, model mismatch, or genuine transport coupling.

## 14 Conclusion

Null leakage is the CTMT mechanism by which unresolved observable variability enters resolved reconstruction through covariance geometry. It is measured by  $C_{RN}$ , constrained by admissible automorphisms, amplified through Schur complements, tested through Fisher conditioning, and globalized through holonomy and atlas overlap defects.

The final message is precise. Leakage is not failure. Leakage is openness of the observable covariance geometry. Instability occurs only when that openness breaks a required health bar: rank, sectors, covariance, coupling, Fisher response, holonomy, or atlas gluing. This gives CTMT a falsifiable and operational language for distinguishing stable openness from collapse.

## A Appendix A: JSON Output Schema

```
{
  "rank": 4,
  "P_R": [...],
  "P_N": [...],
  "Sigma_R": [...],
  "C_RN": [...],
  "Sigma_N": [...],
  "leakage_norm": 0.0245,
  "normalized_leakage": 0.12,
  "Schur_resolved": [...],
  "cond_Schur": 15.2,
  "Fisher": [...],
  "cond_Fisher": 18.7,
  "automorphism_tests": [...],
  "holonomy_leakage_errors": [...],
  "atlas_overlap_leakage_errors": [...],
  "instability_declared": false
}
```

## B Appendix B: Minimal Pseudocode

```
U, s, Vt = svd(J, full_matrices=True)
rank = count(s > tol)
U_R = U[:, :rank]
U_N = U[:, rank:]
P_R = U_R @ U_R.T
P_N = U_N @ U_N.T
Q = concatenate([U_R, U_N], axis=1)
Sigma_block = Q.T @ Sigma @ Q
Sigma_R = Sigma_block[:rank, :rank]
C_RN = Sigma_block[:rank, rank:]
Sigma_N = Sigma_block[rank:, rank:]
leakage_norm = norm(C_RN, 'fro')
normalized = leakage_norm / sqrt(norm(Sigma_R, 'fro')*norm(Sigma_N, 'fro'))
S_R = Sigma_R - C_RN @ inv(Sigma_N) @ C_RN.T
F = J.T @ inv(Sigma) @ J
for test in tests:
    compute_rank_projector_covariance_coupling_fisher_holonomy_errors(test)
```



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